

Mingjia Huo

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Education

University of California, San Diego (UCSD)

PhD in Electrical and Computer Engineering (GPA: 4.0)

- Advisor: Hao Zhang, Tajana Rosing

California, US

Sep 2023 - June 2027

University of Illinois, Urbana-Champaign (UIUC)

MS in Electrical and Computer Engineering (GPA: 3.95)

- Advisor: Kirill Levchenko

Illinois, US

Aug 2020 - Dec 2022

Peking University

BS in Computer Science (Turing Class, GPA: 3.76, rank top 10%)

- Advisor: Qun Huang

Beijing, China

Sep 2016 - Jun 2020

Research Projects

Action-Conditioned World Model Training

Jan 2026 - June 2026

- Added the action module pipeline on top of Wan2.1-1.3B video generation model, fine-tuned on 250K game Zelda videos to achieve consistent action following while preserving visual fidelity.
- Distilled the 40-step bidirectional teacher model into a 4-step causal student model using diffusion forcing and self-forcing, with an adaptive forcing strategy that progressively extends student rollout under dynamic teacher guidance.
- Improve performance on out-of-distribution scenes by scaling action embeddings and attention-map injection mechanisms.
- Optimized inference latency with DeepSpeed Ulysses sequence parallelism to reduce inference latency.

Imgame Bench and Multi-Turn RL Training on Games

Feb 2025 - Oct 2025

- Benchmarked state-of-the-art LLMs on six video games under various settings, both with and without structured harnesses to aid game-state understanding, and established Imgame bench leaderboard.
- Built LLM agents with visual perception and memory modules for interacting with game environments.
- Led multi-turn RL training of Qwen2.5-7B-Instruct on text-only Sokoban and Tetris, and demonstrated improved generalization on spatial reasoning, planning and multi-turn decision-making tasks.

LLM Watermarking

Sep 2023 - Feb 2024

- Applied logit-based LLM watermarking and trained adaptive controllers to control watermark strength from token history.
- Designed a multi-objective optimization framework to balance detectability and semantic coherence.
- Evaluated on C4 dataset and demonstrated improvements on the Pareto frontier of detection-semantics trade-off curves.

Publication

[1] Mingjia Huo, Yao Fu, Bo Chang, et al. Evaluating Spatiotemporal Reasoning of Vision-Language Models in Atari Gameplay. Submitted.

[2] Mingjia Huo, Kaiqin Kong, Minshen Zhang, et al. Robust Action-Conditioned World Modeling for Third-Person Games. Submitted.

[3] Lanxiang Hu, Mingjia Huo, Yuxuan Zhang, et al. Imgame-Bench: How Good are LLMs at Playing Games? ICLR'26.

[4] Mingjia Huo, Sai Ashish Somayajula, Youwei Liang, et al. Token-Specific Watermarking with Enhanced Detectability and Semantic Coherence for Large Language Models. ICML'24.

Work Experience

MBZUAI - Institute of Foundation Models

California, US

AI Research Intern

Jun 2026 - Present

- Trained action-conditioned video generation models on diverse action and prompt sets to enable third-person character-environment interactions.
- Developed a scalable game HUD/UI removal pipeline with VLM-based bbox annotation, YOLO detector training, temporal bbox post-processing, and DiffuEraser/Omni inpainting.
- Applied DMD to distill multi-step causal video generation models into few-step models for real-time inference.

Google DeepMind - Gen AI

California, US

Student Researcher

Jun 2025 - Oct 2025

- Developed a benchmark based on Atari games to evaluate high-FPS video understanding and decision-making of VLMs.
- Designed an agentic workflow to dynamically adjust context history of game-play.
- Identified key failure modes of VLMs on motion detection, spatiotemporal reasoning and long context reasoning.

Skills

Tool PyTorch, Matlab, Kubeflow, Snowflake, Adobe Illustrator
Programming Python(Fluent), SQL(Fluent), C, C++

University Work Experience

2025	Teaching Assistant, DSC 204A: Scalable Data Systems	UCSD
2025	Teaching Assistant, ECE 268: Security of Hardware Embedded Systems	UCSD

Selected Awards

2019	Fellowship , Hui-Chun Chin and Tsung-Dao Lee Chinese Undergrad Research Endowment
2015	Silver Medal , Chinese Mathematical Olympiad
2015	Gold Medal , Chinese Girls' Mathematical Olympiad